

Embedded Figures

COMPLETE LEARNING GUIDE

What You'll Learn

Embedded (or 'hidden') figure questions show a simple target shape — often a specific triangle, quadrilateral, or a combination like a triangle overlapping a rectangle — and ask which of several larger, more cluttered figures contains that exact shape hidden within its lines. The complex figures are deliberately drawn with extra crossing lines, extra shapes, and near-miss outlines so that a shape which looks 'close enough' at a glance often differs in one key detail — an extra line, a missing vertex, a slightly different angle. The reliable method is to reduce the target shape to its defining, checkable features (number of sides, whether any side is noticeably longer/shorter than the others, whether an angle is roughly a right angle or acute) and then trace that specific combination through the complex figure's lines only, ignoring every line that isn't needed. This sub-module builds that feature-checklist method with fully worked examples.

1. Reducing the Target Shape to a Checklist

Before scanning any complex figure, describe the target shape in words as a short checklist: number of sides, any side that is clearly the longest or shortest, whether it contains a right angle, and its rough overall proportion (tall and narrow, wide and flat, roughly equal-sided). This checklist becomes the test applied to every candidate figure.

- Target shape: a right-angled triangle where one leg is roughly twice the length of the other. Checklist: 3 sides; exactly one right angle; the two legs forming that right angle are unequal, with one about double the other; the hypotenuse is the longest side. Any embedded triangle failing even one item (e.g., an equilateral triangle, or a right triangle with equal legs) is rejected immediately.

2. Tracing Technique: Follow Only the Needed Lines

Within a complex figure, place a finger (or mentally trace) starting at one plausible vertex of the target shape and follow only the lines that would form its outline, actively ignoring every other line that crosses through — extra lines in embedded-figure puzzles exist purely to distract and are never part of the hidden shape unless they coincide with it.

- A complex figure is a large triangle divided by two internal lines crossing near its center, creating a small quadrilateral in the middle. To check if that central quadrilateral matches a target 'square-ish' shape, trace only its 4 bounding segments — the two internal crossing lines plus two segments of the outer triangle's sides — while ignoring the rest of the outer triangle's outline.

3. Vertex Counting as a Fast Filter

Before fully tracing a candidate, do a fast count: identify how many distinct line intersections (vertices) would be needed to close the target shape, and check whether that many suitably-placed intersection points even exist near each other in the complex figure — if not, that entire region can be eliminated without a detailed trace.

- Target shape needs 4 vertices forming a quadrilateral. A candidate complex figure has a region where only 3 lines intersect near each other (giving at most 3 usable vertices in that area) — that region is eliminated instantly without needing to check angles or side lengths.

4. Watching for Near-Miss Distractors

The most common wrong-answer trap is a shape in the complex figure that matches the target in side COUNT but not in proportions or angle — e.g., a triangle with the right number of sides but drawn noticeably more equilateral, or a quadrilateral that is a parallelogram when the target was specifically a shape with one right angle.

- Target: a quadrilateral with exactly one right angle (like an L-corner shape) and two unequal pairs of sides. A complex figure contains a 4-sided region that LOOKS similar but has two pairs of parallel sides (a parallelogram, no right angle) — despite having 4 sides like the target, it fails the 'right angle' checklist item and must be rejected.

5. Embedded Figures Formed by Overlap

Some target shapes are formed not from a single traced outline but from the overlapping region of two figures (e.g., the shape common to both an oval and a rectangle drawn over each other) — for these, identify the target's outline as the boundary where both original shapes' edges alternate, then search the complex figure for that same alternating boundary pattern.

- Target: the lens-shaped overlap of two circles. Look for a complex figure containing two curved arcs (each a piece of a different circle) meeting at two points to enclose a lens shape — a complex figure with only straight lines cannot contain this target regardless of how similar it looks, since curved boundaries are a non-negotiable feature.

6. Rotated or Mirrored Embedded Figures

The target shape may be embedded in the complex figure at a different rotation or as a mirror image, not necessarily in its originally shown orientation — before rejecting a candidate region, mentally rotate or flip your checklist's expectations (a right angle stays a right angle under rotation; side-length ratios stay the same under rotation and reflection) rather than only searching for the shape in its exact original orientation.

- Target: a right triangle with the right angle at the bottom-left, legs pointing up and right. A complex figure contains a same-sized, same-proportioned right triangle but rotated 90° so the right angle is at the top-left. Since side-length ratios and the right angle are rotation-invariant, this IS a valid match — rejecting it for 'facing the wrong way' would be an error.

7. Worked Full Example: Finding a Star Inside a Web of Lines

Given a target 5-pointed star outline and a complex figure made of several overlapping triangles creating a web of crossing lines, trace only the 10 line segments needed to form a 5-pointed star's outline (5 outer points, 5 inner concave vertices), verifying that each of the 10 required segments exists as an unbroken line in the complex figure — if even one segment is missing or interrupted by a gap, that candidate fails.

- A complex figure shows 3 overlapping triangles arranged so their edges cross and form a small 5-pointed star shape at the center, exactly where all 3 triangles overlap. Tracing the star's 10-segment outline confirms every segment is a genuine unbroken line from one of the 3 triangles — this candidate is a valid embedded figure. A second candidate figure has the same star outline but one of its 5 outer points is formed by a gap (a dotted, not solid, line) — this candidate is rejected.

Common Pitfalls

- Accepting a shape that matches the target's side count but not its proportions or angles (a near-miss distractor).
- Getting distracted by extra lines in the complex figure that aren't part of the traced target outline.
- Rejecting a valid match only because it appears rotated or mirrored relative to the target's shown orientation.
- Skipping the checklist step and judging matches purely by overall visual impression.
- Missing a target shape formed by the overlap/boundary of two different figures rather than a single traced outline.
- Not verifying that every required segment is a continuous, unbroken line — accepting a shape with a gap or dotted segment.
- Spending equal time on every candidate instead of using the fast vertex-count filter to eliminate obviously wrong regions first.

Key Takeaways

- Reduce the target shape to a short checklist (side count, longest/shortest side, right angle or not) before scanning any candidate.
- Trace only the lines needed for the target outline; ignore every other line in the complex figure.
- Use a quick vertex/intersection count to eliminate regions that can't possibly form the target before doing a detailed trace.
- A valid match may be rotated or mirrored from the target's shown orientation — angle and side-ratio checks still apply.
- Reject near-miss shapes that match side count but fail on angle type or side-length ratio.

- For overlap-based targets, look for the correct alternating-boundary pattern, including curved edges if the target has them.

Learn More

IndiaBix — Non-Verbal Reasoning: Spotting Embedded Figures

<https://www.indiabix.com/non-verbal-reasoning/spotting-embedded-figures/>

GeeksforGeeks — Non-Verbal Reasoning Questions and Answers

<https://www.geeksforgeeks.org/non-verbal-reasoning-questions-and-answers/>